

One-bit DOA estimation based on generalised expectation maximisation sparse Bayesian learning

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Abstract—One-bit quantisation is an appealing low resolution acquisition strategy for large scale array systems, as it significantly reduces the hardware cost and data rate. However, the resulting nonlinear distortion makes it challenging to infer both the number of sources and their directions of arrival (DOAs) from one-bit measurements. This paper proposes a generalised expectation maximisation sparse Bayesian learning framework (GEM-SBL) for one-bit state estimation. By leveraging the arcsine law, the proposed method constructs second order statistics from quantised observations and then performs covariance domain SBL within a GEM procedure. Simulation results, evaluated using the generalised optimal sub pattern assignment (GOSPA) metric, demonstrate that the proposed approach achieves reliable performance across a wide SNR range.

Index Terms—one-bit quantisation, DOA estimation, sparse Bayesian learning, generalised EM.

I. INTRODUCTION

Direction-of-arrival (DOA) estimation with sensor arrays has long been a central topic in array signal processing, due to its broad applications in passive sensing and electronic warfare systems [1]–[4]. For DOA estimation and tracking, a number of mature approaches have been developed, such as variational Bayesian (VB) based estimation methods [5], [6], tensor decomposition based tracking techniques [7], and direct tracking using the generalized labeled multi-Bernoulli filter [8]. These methods have been shown to effectively address the DOA detection and tracking problem.

In practical systems, however, the acquired array data are often subject to severe hardware and link constraints, e.g., limited ADC resolution, stringent power budgets, and restricted fronthaul capacity [9]–[11]. Among various low resolution architectures, one-bit quantisation represents an attractive choice because it can substantially simplify the receiver front end and reduce the data rate [12], [13]. Recovering the number of sources and their DOAs from one-bit measurements is therefore an important and challenging problem.

One-bit quantisation introduces severe nonlinear distortion that fundamentally alters the structure of the received array data. In particular, when the SNR is low, the quantised

observations become highly noisy, and it is considerably more difficult to extract reliable directional information [14], [15]. Moreover, reconstructing the original unquantised measurements from one-bit data is infeasible, since the absolute amplitude and fine grained phase information are irreversibly lost [16], [17]. As a consequence, methods that critically depend on the first order characteristics of the unquantised signal model—e.g., conventional VB formulations built upon a Gaussian snapshot likelihood—become difficult to implement directly in the one-bit setting.

A variety of approaches have been developed for one-bit DOA estimation. A popular line of work exploits the arcsine relationship to reconstruct second order statistics and then applies subspace based estimators, leading to one-bit MUSIC solutions [18]. In parallel, sparse recovery and Bayesian inference formulations have also been investigated, including compressed sensing based algorithms and one-bit sparse Bayesian learning methods, which can incorporate sparsity and offer improved robustness in complex environments [19]. Despite these advances, many existing methods rely on an accurate estimate of the source number; in complex environments, this estimate can be highly unreliable, which in turn severely degrades the overall performance.

Accordingly, this paper proposes a one-bit state estimation method, i.e., joint inference of the source number and DOAs, based on a generalised expectation maximisation [20] sparse Bayesian learning (GEM-SBL) framework. Specifically, within the GEM procedure, we exploit the arcsine law [21] in the E-step to construct second order statistics from one-bit measurements, and then, in the M-step, estimate the states using a covariance based SBL algorithm. In the simulations, we adopt the generalised optimal sub pattern assignment [22] (GOSPA) metric for performance evaluation and compare the proposed method with the covariance driven one-bit MUSIC baseline [18], followed by a detailed analysis of the results. The proposed algorithm performs reliably across all SNR levels.

The remainder of the paper is organised as follows. Section II introduces the signal model and the prior distributions. Section III presents the GEM method and the VB updates for latent variables. Section IV proposes an algorithm based

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on GEM-SBL and elaborates on the initialisation details. Section V presents extensive simulation results and comparisons with baselines. Finally, Section VI summarises the findings and outlines potential extensions.

II. BACKGROUND

A. Problem formulation

Consider a multiple signal receiving scenario with a uniform linear array. Supposing that there are M_t sources at time t . Since M_t is unknown for receiver, we assume that there are K potential sources (PSs) ($K \geq M_t$). Each potential source is associated with a unique DOA $\theta_{i,t}$, which corresponds to an array manifold given by

$$a(\theta_{i,t}) \triangleq [1, e^{-jm \sin(\theta_{i,t})}, \dots, e^{-jm(L-1) \sin(\theta_{i,t})}]^\top, \quad (1)$$

where $m = \frac{2\pi d}{\lambda}$ with d denoting the uniform antenna spacing, λ the signal wavelength, and L the number of antenna elements in the array.

Then the measurement Y_t (a $L \times N$ matrix) can be modeled as

$$Y_t = \sum_i^K a(\theta_{i,t}) x_{i,t} + V_t, \quad (2)$$

where $x_{i,t}$ is a row vector of length N containing the signal samples and V_t denotes the white Gaussian noise with zero mean. It should be noted that for PS which remain inactive at time t , we have $x_{i,t} = \mathbf{0}_{1 \times N}$.

Then, we can write a concise expression of measurement

$$Y_t = A(\Theta_t) X_t + V_t, \quad (3)$$

where $\Theta_t \triangleq [\theta_{1,t}, \dots, \theta_{K,t}]$, $A(\Theta_t) \triangleq [a(\theta_{1,t}), \dots, a(\theta_{K,t})]$ and $X_t \triangleq [x_{1,t}^\top, \dots, x_{K,t}^\top]^\top$. (2)-(3) introduce the measurement model which is unquantified.

In practice, the receiver does not have access to the high resolution Y_t in (3). Due to power and data rate constraints, each antenna element is equipped with a one-bit ADC, which only preserves the signs of the in phase and quadrature components.

We use Z_t to denote the measurement after one-bit sampling, is obtained by applying a quantisation function $Q(\cdot)$ to Y_t , i.e.,

$$Z_t = Q(Y_t). \quad (4)$$

Here, Q operates element wise on the matrix Y_t . Specifically, for each entry (m, n) ,

$$Q(Y_t)_{(m,n)} = \text{sign}\{\Re(Y_t)_{(m,n)}\} + j \cdot \text{sign}\{\Im(Y_t)_{(m,n)}\}, \quad (5)$$

where j represents the imaginary unit, $\Re(\cdot)$ and $\Im(\cdot)$ extract the real and imaginary parts, respectively.

Therefore, the problem addressed in this paper can be summarised as follows. At time t , the signal received by a uniform linear array (ULA) through the one-bit ADC is $Z_t = Q(Y_t)$, from which we need to estimate the number of active sources and their states.

B. Prior probability distribution

Given the assumed independence across variables, the joint distribution factorised form

$$f(Z, X, \Sigma) = f(Z|X, \Sigma) f(X, \Sigma). \quad (6)$$

Since the exact likelihood $f(Z|X, \Sigma)$ is analytically intractable due to the one-bit quantisation, we instead adopt a GEM-SBL framework to obtain a tractable surrogate for inference.

Then, the joint distribution can be written as

$$f(Z, \mathcal{X}) = f(Z|Y, X, \Sigma) f(Y|X, \Sigma) f(X, \Sigma), \quad (7)$$

where $f(Z|Y, X, \Sigma) = f(Z|Y)$ is the conditional PDF.

Specifically, in the E-step we construct the required first and second order moments associated with the quantisation model $f(Z|Y)$. And in the M-step, given these statistics, we perform VB to update the variational posteriors and iteratively estimate the unknown parameters in $\mathcal{X}$.

The following will introduces the PDF based on $f(Y, X, \Sigma)$ and related content. Similar to (7), the joint distribution based on Y is

$$f(Y, X, \Sigma) = f(Y|X, \Sigma) f(X|\Sigma) f(\Sigma). \quad (8)$$

Considering the additive complex Gaussian noise of variance ν , the measurement likelihood is

$$f(Y|X, \Sigma) = f_{\text{MCN}}(Y; A(\Theta)X, \nu I), \quad (9)$$

where $f_{\text{MCN}}(\cdot; Z_c, \Sigma_c)$ denotes the matrix complex Gaussian distribution with mean Z_c and row covariance Σ_c .

Here we adopt an SBL discretisation: we consider K PSs and associate each PS with a unique grid angle. During the subsequent iterations, the estimated powers of PS whose angle does not match the true DOA will shrink to zero via automatic relevance determination (ARD) [23]. When updating the remaining latent variables, the grid angles are treated as known within each iteration.

To facilitate sparse inference, we assign each coefficient x_i a zero-mean circularly symmetric complex Gaussian prior, and place an independent inverse-Gamma (IG) hyperprior on the corresponding variance σ_i (the i -th diagonal entry of Σ). Under the assumption that the coefficients associated with different grid angles are mutually independent, the joint prior factorises as

$$f(X, \Sigma) = \prod_{i \in \mathcal{K}} f(x_i | \sigma_i) f(\sigma_i), \quad (10)$$

$$f(x_i | \sigma_i) = f_{\text{CN}}(x_i; 0, \sigma_i), \quad (11)$$

$$f(\sigma_i) = f_{\text{IG}}(\sigma_i; \alpha_i, \beta_i), \quad (12)$$

where $\mathcal{K} \triangleq \{1, 2, \dots, K\}$ is the index set. Here, $f_{\text{CN}}(\cdot; \mu, \Sigma)$ denotes the complex Gaussian distribution with mean μ and covariance Σ , and $f_{\text{IG}}(\cdot; \alpha, \beta)$ denotes the IG distribution with shape parameter α and scale parameter β . This hierarchical construction enables irrelevant components to be suppressed during the iterations and thereby reveals the active sources.

Accordingly, in our setting, ν is provided by external calibration and kept fixed throughout the iterations; hence, we omit the update of ν .

III. DERIVATION

For ease of expression, we temporarily drop the time index t without leading to any confusion.

A. E-step

In the proposed GEM framework, the E-step aims to construct tractable statistics of the latent unquantised measurements Y corresponding to the one-bit data Z . Given the current estimates of (X, Σ) from the previous M-step, we seek the conditional moments

$$M_Y \triangleq \mathbb{E}[Y|Z, X, \Sigma], \quad S_Y \triangleq \mathbb{E}[YY^H|Z, X, \Sigma], \quad (13)$$

where M_Y and S_Y denote the first and second order moments, respectively. Since a closed form evaluation of (13) under one-bit quantisation is generally intractable, we approximate the required second order moment by a covariance domain surrogate obtained via the arcsine law. Let

$$\hat{R}_Z \triangleq \frac{1}{N} Z Z^H \quad (14)$$

be the sample covariance of the one-bit measurements. According to the arcsine law, the covariance of Z is related to that of Y through an element-wise nonlinear mapping, i.e.,

$$R_Z = \mathcal{A}(R_Y), \quad (15)$$

where $R_Y = \mathbb{E}[YY^H]$ and $\mathcal{A}(\cdot)$ denotes the arcsine-law mapping. Consequently, we recover an estimate of the unquantized covariance by the corresponding inverse mapping,

$$\hat{R}_Y \approx A^{-1}(\hat{R}_Z), \quad (16)$$

which serves as the second order statistic required by the subsequent M-step.

B. M-step

In the M-step, conditioned on the statistics obtained in the E-step, we perform VB to update the posterior distributions of X and Σ . Specifically, under the mean field theory, all posterior variables are assumed to be mutually independent. Incorporating this independence assumption yields

$$q(X, \Sigma) = q(X) \prod_{i \in \mathcal{K}} [q(\sigma_i)], \quad (17)$$

which allows for the alternative performing optimisation on the specific distributions.

1) *Computing $q^{(p)}(X)$* : Combining (8) and (17), the variational posterior distribution of X can be written as

$$q^{(p)}(X) \propto \exp \left\{ -\frac{1}{\nu} \mathbb{E}_{\sim X} [\|Y - A(\Theta)X\|_F^2] - \|X\|_F^2 (\hat{\Sigma}^{(p-1)})^{-1} \right\}, \quad (18)$$

where $\mathbb{E}_{\sim X}(\cdot)$ is the expectation of variables except X . $\hat{X}$ and $\hat{\Sigma}$ respectively denote the mean matrix and row covariance matrix of the posterior $q(X)$, which is defined as

$$q(X) = f_{\text{CN}}(X; \hat{X}, \hat{\Sigma}). \quad (19)$$

In standard VB method, we need to expand (18) based on (19), which yields a closed form updates for $\hat{X}$ and $\hat{\Sigma}$

$$\hat{\Sigma}^{(p)} = \left(\text{diag} \left(\frac{u^{(p-1)}}{v^{(p-1)}} \right) + \frac{1}{\nu} (A^H A) \right)^{-1}, \quad (20)$$

$$\hat{X}^{(p)} = \frac{1}{\nu} \hat{\Sigma}^{(p-1)} A^H Y, \quad (21)$$

where

$$u^{(p-1)} = [\hat{\alpha}_1^{(p-1)}, \dots, \hat{\alpha}_K^{(p-1)}], \quad (22)$$

$$v^{(p-1)} = [\hat{\beta}_1^{(p-1)}, \dots, \hat{\beta}_K^{(p-1)}], \quad (23)$$

collect the IG hyper-parameters of σ_i , which is updated in the following section.

Noted that the update of $\hat{X}$ in (21) depends on the unquantised measurement Y , which is unavailable under one-bit sampling. To circumvent this issue, we consider the second order statistic $W \triangleq \mathbb{E}[XX^H]$. Using (21), we have

$$\widehat{W}^{(p)} = \frac{\hat{\Sigma}^{(p-1)} A^H \hat{R}_Y A (\hat{\Sigma}^{(p-1)})^H}{\nu^2} + \hat{\Sigma}^{(p-1)} \quad (24)$$

Equations (20) and (24) enable covariance driven updates of $\hat{\Sigma}$ and $\widehat{W}$ using the surrogate statistic $\hat{R}_Y$ obtained in the E-step; the remaining updates are then derived based on these quantities.

2) *Computing $q^{(p)}(\sigma_i)$* : Combining (8) with (10), the variational factor associated with σ_i can be written as

$$q^{(p)}(\sigma_i) \propto f(\sigma_i) \exp \left\{ \int \ln f(x_i | \sigma_i) q^{(p-1)}(x_i) dx_i \right\}. \quad (25)$$

With an IG prior imposed on σ_i , conjugacy ensures that $q^{(p)}(\sigma_i)$ remains IG. Consequently, the shape and scale parameters are updated as

$$\hat{\alpha}_i^{(p)} = \hat{\alpha}_i^{(0)} + N, \quad (26)$$

$$\hat{\beta}_i^{(p)} = \hat{\beta}_i^{(0)} + \mathbb{E} \left[x_i^{(p-1)} (x_i^{(p-1)})^H \right]. \quad (27)$$

Here, N denotes the number of temporal snapshots of x_i . The expectation in (27) is obtained from the second order statistic of X , i.e.,

$$\mathbb{E} \left[x_i^{(p-1)} (x_i^{(p-1)})^H \right] = \widehat{W}_{ii}^{(p-1)},$$

where $\widehat{W}^{(p-1)}$ denotes the posterior second moment matrix of X at iteration $p-1$, which is computed according to (24).

IV. PROPOSED METHOD

The complete inter temporal method is proposed in this section, as summarised in Algorithm 1, where the initialisations of priors at each time t are carried out as follows.

- For Θ , we construct a predefined angular grid over $[\theta_{\min}, \theta_{\max}]$ with resolution Δ_θ , and form the dictionary $A = [a(\theta_1), \dots, a(\theta_K)]$. Each PS is associated with one grid angle θ_i .
- Since W is updated first in every iteration, no explicit initial value is required. Specifically, the algorithm directly

- computes $\widehat{W}^{(1)}$ from the current values of the remaining variables.
- For Σ , $q^{(0)}(\sigma_i) = f_{\text{IG}}(\sigma_i; \alpha_i^{(0)}, \beta_i^{(0)})$, where $\alpha_i^{(0)} = 1e^{-3}$ and $\beta_i^{(0)} = 1e^{-3}$.

Algorithm 1 The proposed algorithm

Input: $\widehat{R}_Z$, grid $\Theta = \{\theta_k\}_{k=1}^K$

Output: $\widehat{\Theta}$

- 1: Reconstruct $\widehat{R}_Y$ via (16)
 - 2: Initialise posteriors as in Section III-B
 - 3: **for** $p = 1$ to N_{iter} (maximum number of iterations) **do**
 - 4: Update $q^{(p)}(\Sigma)$ via (26)–(27)
 - 5: Update $q^{(p)}(W)$ via (20)–(24)
 - 6: **if** $\|\widehat{\Sigma}^{(p)} - \widehat{\Sigma}^{(p-1)}\|_F^2 / \|\widehat{\Sigma}^{(p)}\|_F^2 < r$ **then**
 - 7: **break**
 - 8: **end if**
 - 9: **end for**
 - 10: Estimate DOAs by peak picking on the ARD spectrum derived from $\widehat{\Sigma}$
 - 11: Output $\widehat{\Theta}$
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V. NUMERICAL RESULTS

A. Scenario

In this section, we evaluate the DOA estimation performance for 4 sources using a ULA with 10 elements and inter element spacing $d = 1.5$ m. The carrier frequency is set to 100 MHz. The true DOAs are fixed at $[-48^\circ, -17^\circ, 22^\circ, 63^\circ]$. For the SBL based initialisation, the angular sector $[-89^\circ, 89^\circ]$ is uniformly discretized into 179 grid points, the convergence threshold is set to $r = 10^{-3}$, and the maximum number of iterations N_{iter} is set to 50.

We compare the proposed method with the one-bit MUSIC baseline in [18] from two perspectives: noise robustness versus SNR, and information dependence versus the number of snapshots. For a fair comparison, one-bit MUSIC uses the same 179 point grid, and the number of sources is selected by AIC [24]. In contrast, the proposed method does not require the source number as an input.

Quantitative performance is assessed using the GOSPA metric [22]. Given two finite sets X and Y , the GOSPA distance is defined as

$$\text{GOSPA}(X, Y) = \left(\min_{\xi} \sum_{(x, y) \in \xi} \min\{c, \|x - y\|\}^p + \frac{c^p}{\alpha} (|X| + |Y| - 2|\xi|) \right)^{1/p},$$

where ξ denotes a partial matching between elements of X and Y , p is the order of the norm, c is the cut-off parameter, and α controls the trade off between localisation errors and cardinality mismatch. In all simulations, we set $p = 1$, $c = 1$, and $\alpha = 1$.

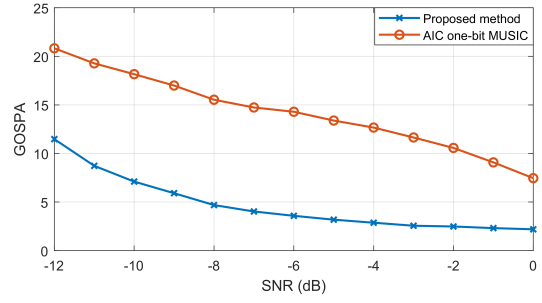


Fig. 1: DOA Performance with SNR (with snapshots = 100)

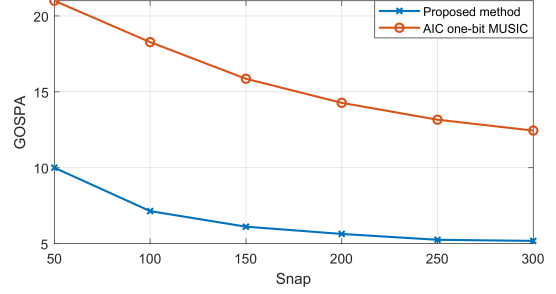


Fig. 2: DOA Performance with Snapshot (with SNR = -10 dB)

B. Performance evaluation

First, we evaluate the performance of algorithms under different SNR levels, with the number of snapshots fixed to 100. As shown in Fig. 1, the proposed method consistently outperforms the AIC based one-bit MUSIC over the entire SNR range. This demonstrates improved robustness when the source number is unknown and must be inferred from the one-bit measurements.

Next, to investigate the impact of the available information, we fix the SNR to -10 dB and evaluate the performance versus the number of snapshots, as depicted in Fig. 2. It can be observed that the proposed method achieves lower GOSPA than the AIC based one-bit MUSIC for all snapshot numbers, demonstrating its superiority in the low information regime.

VI. CONCLUSION

This paper investigated one-bit direction-of-arrival estimation under practical hardware and link constraints and proposed a generalised expectation maximisation sparse Bayesian learning (GEM-SBL) framework for joint state inference from one-bit array measurements. By combining arcsine law based second order statistics with covariance domain SBL, the proposed method provides robust estimation in challenging regimes. Simulation results based on the generalised optimal sub pattern assignment metric showed that the proposed approach performs reliably across different SNR levels. Future work will consider extensions to off grid modelling, correlated snapshots, and more general array geometries, as well as further improvements in computational efficiency.

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